

Lecture 5. Probability Theory

The statistical tests that we'll learn about in the next two years cannot tell us whether one variable *really* has an effect on another in the real world. Instead, we can ask questions like:

- “What is the likelihood of finding such a big difference between my two experimental groups by random chance alone?”, or
- “What is the likelihood of these two variables looking like they are associated by random chance alone?”.

If the probability is low, then it is highly **unlikely** that the difference between groups (or association between variables) is only due to random variations in the data. Such a low probability is a good thing for us as researchers – it means that the difference/association could be due to a real effect. This is why probabilities are essential for inferential hypothesis testing – they allow us to determine whether our experimental results are too surprising (unlikely) to exist if there was *not* a real effect.

Probabilities

Probability is a number from 0 (impossibility) to 1 (certainty) – the bigger the number, the more likely the event.

$$p = \frac{\text{The number of ways the event could occur}}{\text{The number of possible outcomes}}$$

Example: if I give you a multiple-choice question with 4 possible answers, where only 1 of the 4 is correct, the probability of guessing the correct answer at random is $p = \frac{1 \text{ correct answer}}{4 \text{ possible answers}} = \frac{1}{4} = 0.25$, or 25%.

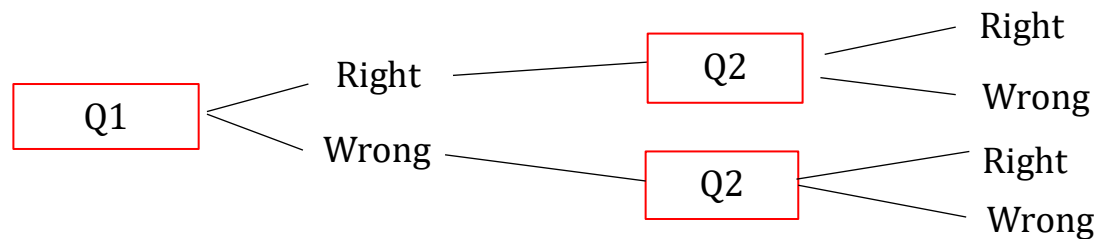
Key terms:

- **Experiment** = activity that produces or observes a new outcome (e.g., selecting an answer, flipping a coin)
- **Sample space** = set of possible outcomes (e.g., {correct:incorrect} or {heads:tails})
- **Elementary event** = one outcome within the sample space (e.g., “correct” or “heads”)

Example experiment:

Let's imagine that our **experiment** consists of selecting the answer to **two** multiple choice questions are random.

- The **event** we are interested in is whether each answer is correct
- The **sample space** now has 4 outcomes in it {RightWrong: RightRight: WrongWrong, WrongRight}
- An **elementary event** is any one outcome, e.g. RightRight



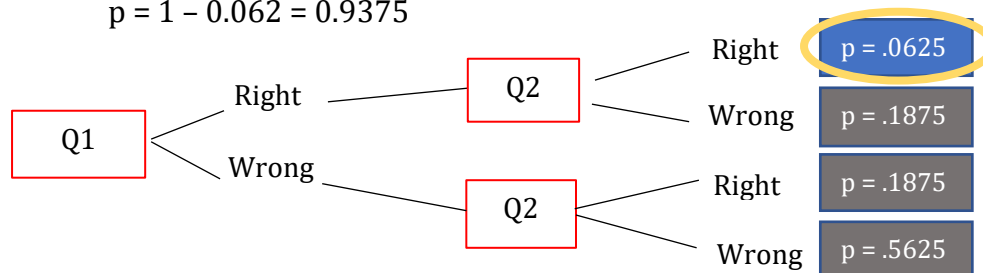
Rule 1: Probability must be between 0 and 1
For any event A, $0 \leq P(A) \leq 1$

Rule 2: The total probability of all outcomes in the sample space adds up to 1.

Rule 3 (rule of subtraction): The probabilities of A happening and *not* happening add up to 1. In other words, the probability of A *not* happening is 1 minus the probability of it happening.

$$P(\bar{A}) = 1 - P(A)$$

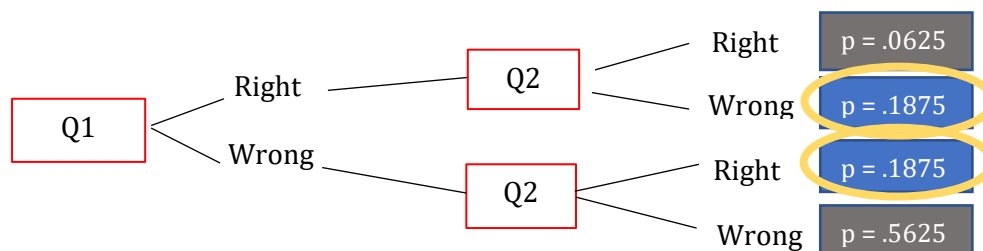
For example, if the probability of getting both answers correct is $p=0.062$, then the probability of *not* getting both answers correct is $p = 1 - 0.062 = 0.9375$



Rule 4 (rule of addition): The probability of *either one* of several events happening is the added probability of each individual event.

$$P(A \cup B) = P(A) + P(B)$$

("A $\cup$ B" means "A **or** B **or** both occurring")

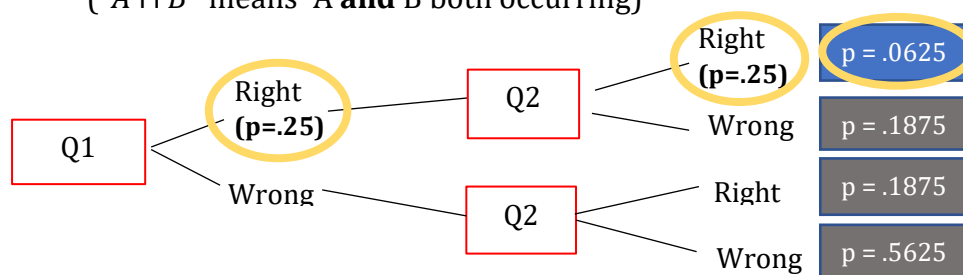


$$P(\text{RightWrong} \cup \text{WrongRight}) = .1875 + .1875 = .375, \text{ or } 37.5\%$$

Note: This only applies to *mutually exclusive* events, i.e., they cannot happen at the same time. For cases where the events are not mutually exclusive, the calculation is slightly different:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

("A $\cap$ B" means "A **and** B both occurring")



$$P(A \cup B) = .25 + .25 - .0625 = .4375$$

Rule 5 (rule of multiplication):

If A and B are independent of each other:

$$P(A \cap B) = P(A) * P(B)$$

So, correct answer on both questions = .25 * .25 = .0625

Probability as Frequency

What is probability?

- **Bayesian approach:** probability is degree of **belief** in an event:
"I believe that there is a 10% chance of selecting a left-handed person based on my past experience of how many people I know who are left-handed"
- **Frequentist approach:** probability is the long run **frequency** of an event if **repeated** many times:
"To determine probability, I need to select you at random and ask whether you are left-handed, and work out what the frequency of left-handers is."

From a frequentist approach, we can determine the probability of outcomes of an experiment by repeating the experiment many times and tallying the frequency of each event in the sample space.

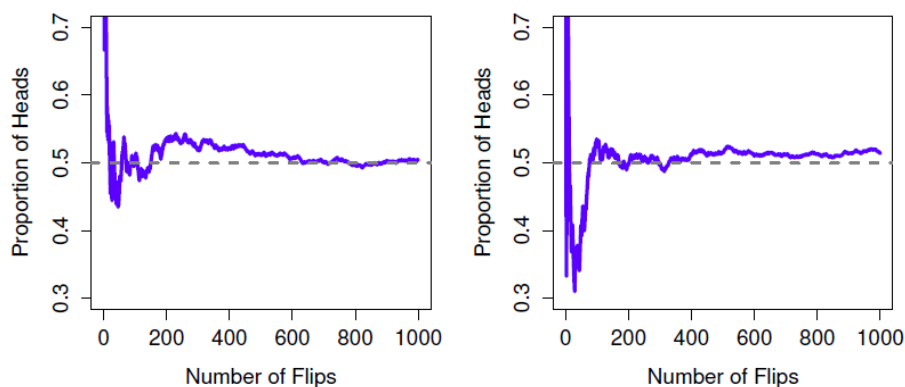
Back to our example with a single multiple-choice question (1 right answer, 3 wrong answers). We can run this experiment 10 times in R, using a random number generator. The results are represented in the table on the right.

Experiment	Outcome
1	Right
2	Right
3	Wrong
4	Wrong
5	Right
6	Wrong
7	Wrong
8	Wrong
9	Wrong
10	Wrong

We got the right answer 3 times out of 10, so the empirical probability is $\frac{3}{10} = \mathbf{0.33}$, whereas we know that the true probability is $\frac{1}{4} = \mathbf{0.25}$. In other words, when we run too few trials, the empirical probability we observe might differ significantly from the real probability.

To remedy this, we can run the same experiments many more times.

After running 1000 trials, we find that the correct answer was selected 241 times. The empirical probability is now $\frac{241}{1000} = .241$ – still not exactly .25, but closer. This is an example of **the law of large numbers**: The empirical probability will approach the true probability as the sample size increases. It implies that the results with small samples can be far off – the larger the dataset we have, the closer our data will be to the truth.



Theoretical probability is determined mathematically and is based on assumptions (e.g., a coin toss assumes the coin is fair, so p is exactly 50:50; or that there is an equal chance of picking any answer to the multiple choice question).

In statistics, we frequently use theoretical probability **distributions** to calculate probabilities. The distribution we use depends on the type of data we are collecting, but two common distributions are:

The binomial distribution

The binomial distribution gives the **discrete** (rather than continuous) probability distribution of obtaining exactly N “successes” on X number of trials which each have two possible outcomes. The exact shape of the distribution is determined by 2 pieces of information: (1) the probability of a “success” on a single trial, and (2) the number of trials in the experiment.

Example: Number of questions correct in 3 question multiple choice quiz where there are 4 response options to each question, and answers chosen are at random

- **experiment:** picking 3 responses at random
- **sample space:** {0 correct, 1 correct, 2 correct, 3 correct}
- **elementary event:** e.g., 3 answers correct

Quiz:		
Q1:	Q2:	Q3:
A: ✗	A: ✓	A: ✗
B: ✓	B: ✗	B: ✗
C: ✗	C: ✗	C: ✗
D: ✗	D: ✗	D: ✓

Calculating the probability of each elementary event:

First	Second	Third	Event	p		Event	P
Wrong	Wrong	Wrong	0	$0.75 \times 0.75 \times 0.75$	0.422	0	0.422
Wrong	Wrong	Right	1	$0.75 \times 0.75 \times 0.25$	0.141	1	0.423
Wrong	Right	Wrong	1	$0.75 \times 0.25 \times 0.75$	0.141	2	0.141
Wrong	Right	Right	2	$0.75 \times 0.25 \times 0.25$	0.047	3	0.016
Right	Wrong	Wrong	1	$0.25 \times 0.75 \times 0.75$	0.141		
Right	Wrong	Right	2	$0.25 \times 0.75 \times 0.25$	0.047		
Right	Right	Wrong	2	$0.25 \times 0.25 \times 0.75$	0.047		
Right	Right	Right	3	$0.25 \times 0.25 \times 0.25$	0.016		

Calculating probabilities in R

What is the probability of getting no answers correct?

```
dbinom(x = 0, size = 3, prob = .25)
```

What is the probability of getting one or fewer correct?

```
pbinom(q = 1, size = 3, prob = .25)
```

What is the probability of picking 2 or more?

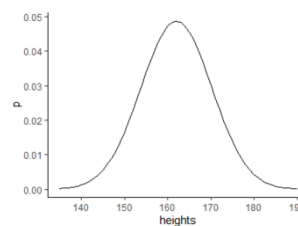
```
pbinom(q = 1, size = 3, prob = .25, lower.tail = FALSE)
```

What number of correct answers has a probability of less than 10%?

```
qbinom(p = 0.1, size = 3, prob = .25, lower.tail = FALSE)
```

The normal distribution

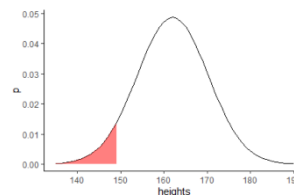
The normal distribution is common when we have **continuous** (rather than discrete) data. Its exact shape is determined by 2 pieces of information: the *mean* and the *standard deviation*.



It doesn't make sense to ask, "What is the probability of being exactly 172 cm tall?" – because height is continuous and no one is *exactly* 172cm tall. Instead, we can look at the probability of scoring higher or lower than a particular point:

```
pnorm(q = 150, mean = 162, sd = 8.18)
```

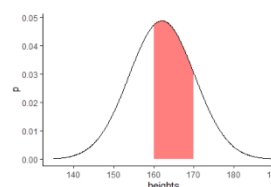
I.e., what is the likelihood of being 150cm or less if the mean height is 162cm (SD=8.18)? The answer is 0.07 – a 7% likelihood.



What is the probability of being between 160 and 170cm?

```
pnorm(q = 170, mean = 162, sd = 8.18) -  
pnorm(q = 160, mean = 162, sd = 8.18)
```

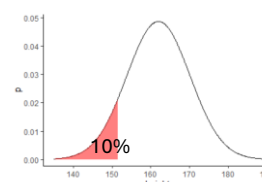
The answer is 0.4325, or 43%.



We can calculate the value that cuts the bottom 10% of the population (e.g., what height do you need to be to be in the shortest 10% of women?)

```
qnorm(p = .10, mean = 162, sd = 8.18)
```

Answer: 151.516, i.e., 10% of women are less than 151.52cm



We can also calculate the value that cuts the top X% of the population using `lower.tail = FALSE`:

```
qnorm(p = .10, mean = 162, sd = 8.18,  
lower.tail = FALSE)
```

